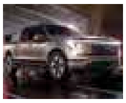
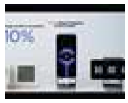


Ford E-Pickup Truck
The F-150 Lightning pickup truck has a base range of 230 miles and 2000 pounds of payload capacity. **http://dgit.in/jun21-33**



Fast Charging at 200W
Xiaomi claims it can fully charge a phone in 8 mins at 200W using its new "HyperCharge" technology. **http://dgit.in/jun21-34**

For keyboard, we have something very intuitive as shown in the example below. We begin by importing pyautogui and time (which we essentially use here to wait for a while). The write() method is used to

write on the screen whatever is written and each key is hit in the interval that is mentioned. The keyDown() function holds the specified key and then the press() function hits the specified key. essentially making it combination. We can also use alert() to display some message within the parameter. Let's take a look at the output. For obtaining the output, we have a notepad file open in which we want the output to be written, which otherwise will let the system type it in the console or in the script file(wherever the cursor points to). It is also for the same reason that we have introduced a sleep for two seconds within which we switch into our notepad file. In the output, we get to see that whatever we had typed in the write() function has been written into the notepad, an alt-tab has occurred (in case of windows, it switches to the other window) and an alert has also appeared. And look at the amount of code that had to be written to perform a relatively simpler automation. Succinct and comprehensible, isn't it? Let's take a look further

Let's use a few of the stuff we learnt above in making something really basic yet interesting.

Remember the dino game chrome offers you to play every time there is limited network connectivity! Let's see if we can do some basic automation here. The logic of the game is pretty simple. We need to allow the dinosaur to run perpetually, making it jump over the cactus or dodge the enemies. Since we aim to keep it basic, let's understand how we can we automate the jump.

For us to play the game, we simply need to hit the space button every time the dinosaur approaches the cactus, which is exactly what we need to do with our code. We need to understand that the area in front of the dinosaur is clear. Note that this area is arbitrary and we get to set the area according to what we feel the best. Essentially, this selection of the area should take into consideration the size of the cactus, which is the obstacle that needs overcoming in this case. That would be structural aspect of the obstacle, but how do we go about identifying the cactus? This is where the colour of the object factors in. When the dinosaur is blazing through, unbridled, it's all white in the screen but the cacti have a dark greyish colour which we can

```
from PIL import ImageGrab
from PIL import ImageOps
import pyautogui
import time
from numpy import *
```

take advantage of. As it stands now, logic dictates that every time we come across an obstacle of dark, greyish colour, we press the space button allowing the dinosaur to jump. Let's take a look into the code now.

To begin with, we need a few packages to make our lives easier. We need the ImageGrab and ImageOps modules from the PIL library(Python Imaging Library).

We will need ImageGrab to take a screenshot of the screen and ImageOps to perform a grayscale conversion, which shall be explained later. Of course, we will need the pyautogui to get hold of the keyboard. We will also need the time module to introduce a certain delay and finally, we import numpy, which is a module that allows us to perform complex mathematical calculations in python.

To begin with, we need to launch the game on the chrome browser. You can either do that by switching off your internet connectivity or by entering chrome://dino in the URL and it should load in this manner

The first thing that we do here is to press the space bar to start the game. Easy enough so far.

The issue becomes complex when we hit an obstacle. We need to restart the game and for that purpose, we need to hit the play button on the screen. One method of obtaining that